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Evaluating the effectiveness of alkylammonium halide-based deep eutectic solvent pretreatments in heightening carbohydrates and lignin separation from poplar sawdust

Wei Tang [a,1], Ruolin Li [b,1], Xinyue Wang [a], Haojie Qian [a], Bo Fan [a], Yucai He [a,*]

[a] School of Pharmacy & School of Biological and Food Engineering, Changzhou University, Changzhou 213164, Jiangsu Province, China
[b] Jiangsu Co-Innovation Center of Efficient Processing and Utilization of Forest Resources, College of Chemical Engineering, Nanjing Forestry University, Nanjing 210037, China

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ABSTRACT

The molecular architecture of hydrogen bond acceptors (HBA) within deep eutectic solvents (DES) emerges as a key factor in bioconversion of lignocellulose. This work constructed a series of DES systems by employing diverse alkylammonium halide as HBAs, enabling an investigation into the synergistic interplay between alkyl chain length and halide anion identity on the deconstruction of poplar sawdust (PS). Notably, the cetyltrimethylammonium bromide (CTAB)-based DES displayed a reduced surface potential, weakened hydrogen bonding energy, and diminished polarity, outperforming counterparts based on short-chain alkylamines or alternative halide-containing HBA. When applied in the pretreatment of PS, this DES has achieved a removal of 68.4 % lignin and 94.0 % xylan, while elevating the cellulose digestibility to 76.5 %. Structural analysis unveiled that the CTAB-based DES profoundly disrupted the pivotal β-O-4 ether bonds and aromatic structure within lignin, and thus enhancing substrate digestibility. The long-chain hydrophobic alkyl chains and highly-polar bromide ions empowered the DES with superior penetration and deconstruction capabilities toward lignocellulose, highlighting the paramount significance of HBA's molecular structure in refining DES pretreatment paradigms for efficient biomass transformation.

1. Introduction

The overreliance on fossil resources and the challenges posed by global climate change necessitates the emergence of renewable resources as a solution to the energy crisis [1]. The bioconversion of abundant and renewable lignocellulosic biomass into green bioenergy is anticipated to facilitate the sustainable development of industrialization [2,3]. *Populus euramericana*, a species extensively cultivated across temperate and cold-temperate zones of the Northern Hemisphere, is prized for its remarkable adaptability and pivotal role in industrial timber production [4]. However, during the processes of harvesting and manufacturing, approximately 30 % of the resulting byproduct manifests as fine particulate matter, commonly known as poplar sawdust (PS) [5]. Much like other lignocellulosic biomass, PS predominantly consists of glucan and xylan, which constitute the fundamental framework of its carbohydrate-rich structure [6]. As such, transforming PS into bioenergy not only presents a promising economic opportunity but also embodies a sustainable and ecologically responsible strategy for waste valorization.

It is widely recognized that lignin, which envelops bioglycans such as cellulose and hemicellulose, forms a complex barrier structure that significantly impedes the biotransformation of lignocellulose [7]. To achieve efficient conversion of wood-based lignocellulose, selecting appropriate delignification pretreatment methods is crucial [8]. Nevertheless, traditional delignification pretreatments always have inherent drawbacks that have limited the development of biorefineries [9]. For instance, the oxidative methods frequently resulted in severe cellulose degradation through over-oxidation [1]. Alkaline processes gave rise to copious amounts of saline effluent [10]. Organic solvent-based systems incurred prohibitively high operational expenses, while ionic liquid treatments were plagued by excessive viscosity and inadequate process efficiency [11]. Oppositely, deep eutectic solvents (DES), which are green and biocompatible mixtures formed by hydrogen bond

* Corresponding author.
E-mail address: heyucai2001@163.com (Y. He).
[1] These authors contributed equally to this work.

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donors (HBD) and acceptors (HBA), have emerged as preferred reaction media for pretreatment [11,12]. The hydrogen bonds between HBA and HBD are the predominant internal non-covalent interactions, playing a crucial role in disrupting the lignocellulosic barrier structure [13]. During the DES pretreatment, the most widely recognized delignification mechanism involved the protons generated from the dissociation of HBD catalyzing the cleavage of ether or ester bonds between lignin and polysaccharide components [14,15]. Furthermore, the HBA would also cleave the Cα site of lignin, subsequently facilitating the cleavage of the β-O-4 bond and generating Hibbert's ketones [16]. For various DES, key properties influencing lignin dissolution included viscosity, pKa, hydrogen bond basicity, and Kamlet-Taft solvatochromic parameters, all closely related to HBA characteristics [17]. Therefore, while numerous studies have concentrated on the relationship between HBD of DES and pretreatment performance, it was undeniable that HBAs also played a crucial role in determining the solubility of lignin in DES [15].

According to the existing literature, DES prepared using HBD primarily comprised organic acids and polyols, whereas HBA predominantly consisted of choline chloride (ChCl) and betaine [14,18]. Regarding pretreatment performance, organic acid-based DES enriched with -COOH and -OH groups exhibited strong correlations with delignification efficiency [19,20]. This was attributed to these organic acids significantly enhancing hemicellulose removal by catalyzing the disintegration of glycosidic bonds and acetyl groups, thereby intensifying the degradation of lignin [5]. For instance, Tian et al. found that formic acid, acetic acid, and lactic acid-based DES achieved delignification extents of 73–77 % and selectivity up to 6 % when treating PS [21]. Hou et al. discovered that the strong electron-withdrawing groups and low pKa of the HBD in acid-based DES facilitated the dissolution of xylan from rice straw [22]. To achieve higher efficiency, acid-based DES can be combined with acid auxiliary agents, such as inorganic acids or metal chlorides, to form ternary DES, thereby ulteriorly improving the depolymerization of xylan and lignin [23,24]. Numerous studies have focused on enhancing the performance of acid-based DES pretreatment by optimizing the acidity or diversity of HBD, while overlooking the potential of alternative HBA beyond quaternary ammonium salts. Ling et al. treated bamboo with DES prepared from different HBA, specifically acetamide, betaine, and ChCl. They discovered that levulinic acid/ChCl exhibited the strongest delignification properties, while levulinic acid/betaine demonstrated superior lignin recovery capabilities [25]. Additionally, it has been reported that a shorter-alkyl-chain in quaternary ammonium salts within DES was more favorable for the dissolution of lignin and xylan, indicating that the selectivity of the HBA significantly influenced pretreatment performance [19]. Our previously works also found that DES prepared by alkylammonium bromide/chloride has unpredictable effects during pretreatments [26]. Specifically, the synergistic effect of cetyltrimethylammonium bromide (CTAB) combined with lactic acid (LA) could achieve the removal of more than 50 % of lignin and 73 % of xylan [27]. This violently disrupted the natural structure of lignocellulose, thereby validating the potential of alkylammonium halide-based DES in biorefinery applications. As the HBA of DES, alkylammonium halides not only exhibited electron-dense domains adept at forming robust hydrogen bonds that promoted efficient eutectic formation, but also boasted extended hydrocarbon chains that conferred pronounced hydrophobicity upon the solvent matrix [28]. This inherent lipophilic character may potentially enhance the extraction efficacy of hydrophobic constituents from lignocellulose. Despite these promising attributes, previous researches have largely been limited to preliminary evaluations of DES pretreatment performance. A deeper mechanistic understanding of how critical structural parameters of HBA (such as the alkyl chain length, the halogen atom type, and the functional group architecture of alkylammonium halides) influenced the overall efficiency of the pretreatment process remained strikingly underexplored. Moreover, the relationship between the structural features of DES's HBA and the deconstruction of lignin required further elucidation. The objective of this study was to investigate the competitive behavior of alkylammonium or ChCl-based DES in the separation of PS components, and to evaluate the potential applications of lower-cost and higher-biocompatibility alkylammonium halides in biorefining processes.

In this study, DES were prepared by combining three types of alkylammonium salts or ChCl with LA. The structural characteristics of the prepared DES were thoroughly analyzed using Fourier-transform infrared spectroscopy (FTIR), proton nuclear magnetic resonance (^{1}H NMR), thermogravimetric analysis (TGA), and Quantum theoretical calculation. Subsequently, these DES were utilized for PS pretreatment, and the cellulose digestibility of the pretreated PS was evaluated. The effects of DES on PS were examined through changes in crystal structure, accessibility, lignin surface area, and lignin structural characterization. Finally, the recyclability of the DES was examined to evaluate their potential application in lignocellulosic pretreatment, and recommendations for optimizing DES pretreatment were provided.

2. Materials and methods

2.1. Materials

The PS was sourced from the Huayuan Wood Processing Plant nestled in Jiangsu Province, China. CTAB, dodecyl trimethylammonium Bromide (DTAB), hexadecyl trimethylammonium chloride (CTAC), and ChCl were procured from Aladdin Biochemical Technology Co., Ltd. LA and other chemical reagents were obtained from Sinopharm Chemical Reagent Group based in Shanghai, China. The commercial cellulase (Cellic CTec2.0) with a filter paper enzyme activity measured at 205 FPU/g was purchased from Sigma-Aldrich's Shanghai branch.

2.2. Preparation and characterization of DES

Various HBA (ChCl, CTAB, DTAB, and CTAC) was mixed with HBD (LA) in 500-mL round-bottomed flasks at molar ratios of 1:2, 1:4, 1:6, and 1:8. The flasks were subsequently sealed and stirred at a constant temperature of 80 °C until transparent and homogeneous solutions were formed. The resulting DES were designated as CCL, CBL, DBL, and CAL, corresponding to the respective HBA used (ChCl, CTAB, DTAB, and CTAC).

The functional groups of DES were investigated at ambient temperature using FTIR spectrometers (Shimadzu, Japan). The instrument was operated at 40 kV and 15 mA, and the FTIR spectra were recorded in the wavenumber range of 4000–500 cm^{-1}. The thermostability of DES was evaluated by TGA using a Rigaku TGA 8122 instrument (Japan) under a nitrogen atmosphere. The TGA measurements were conducted at a constant heating rate of 10 °C/min over a temperature range of 50 °C to 550 °C. Additionally, the proton signals of DES dissolved in DMSO-d_6 were analyzed by ^{1}H NMR spectroscopy using a 400 MHz Bruker AVANCE instrument. The Kamlet-Taft (K-T) parameters were determined based on previous literature using N, N-diethyl-4-nitroaniline, 4-nitroaniline, and Nile red as probes.

2.3. Pretreatment and enzymatic hydrolysis of PS

The PS with a dry weight of 20 g was suspended in diverse DES solutions at a mass ratio of 1:20. The round-bottomed flask containing the mixture was subsequently immersed in an oil bath and stirred at a temperature of 80–140 °C for 60 min. Following this, the pretreated PS and the DES was separated and recovered via vacuum filtration. The pretreated PS was then extensively cycled-washed with hot distilled water (60 °C) to remove the residual DES solvents [29]. After washing, the PS was stored at 4 °C for future use, while the recovered DES was reused for subsequent PS pretreatment according to the aforementioned procedure.

The enzymatic hydrolysis of pretreated PS was conducted in 150-mL conical flasks containing 5 % (w/v) substrate mixed with 50 mL of citric acid buffer solution (50 mM, pH 4.8) and 20 FPU/g-glucan of cellulase.

The flasks were then incubated in a constant temperature shaker at 50 °C and 150 rpm for 72 h. Following incubation, the hydrolysate was centrifuged at 5000*g* for 5 min to obtain the supernatant. The cellulose digestibility of pretreated PS was calculated based on the glucose concentration in the supernatant using Eq. (1).

$$\text{Cellulose digestibility}\ (\%) = \frac{\text{Glucose content of enzymatic hydrolysate}\ (g) \times 0.9}{\text{Glucan content of pretreated PS}\ (g)} \times 100\% \quad (1)$$

2.4. Compositions analysis of PS

The international standard method (NREL) was employed to analyze the chemical composition of raw and pretreated PS. Monosaccharides were quantified via high-performance liquid chromatography (HPLC) with an HPX-87H column manufactured by Agilent Technologies, USA. The lignin content was determined by measuring the weight loss of acid-hydrolyzed residues before and after calcination at 575 °C. The solid/glucan recovery and lignin/xylan removal from PS were assessed according to Eq. (2–3). Furthermore, the separation degree (SD) of lignin (SD_L), xylan (SD_X), and their combined effect (SD_{LX}) was analyzed to assess the efficacy of DES pretreatment to separate xylan and lignin components from PS, as calculated using Eq. (4–6) [7].

$$\text{Recovery of solid or glucan}\ (\%) = \frac{\text{Solid or glucan content of pretreated PS}\ (g)}{\text{Solid or glucan content of raw PS}\ (g)} \times 100\% \quad (2)$$

$$\text{Removal of xylan or lignin}\ (\%) = \left(1 - \frac{\text{Xylan or lignin content of pretreated PS}\ (g)}{\text{Xylan or lignin content of raw PS}\ (g)}\right) \times 100\% \quad (3)$$

$$SD_L(\%) = \frac{\text{Glucan and xylan contents/lignin content of pretreated PS}\ (g)}{\text{Glucan and xylan contents/lignin content of raw PS}\ (g)} \times 100\% \quad (4)$$

$$SD_X(\%) = \frac{\text{Glucan and lignin contents/xylan content of pretreated PS}\ (g)}{\text{Glucan and lignin contents/xylan content of raw PS}\ (g)} \times 100\% \quad (5)$$

$$SD_{LX}(\%) = \frac{\text{Glucan content/xylan and lignin contents of pretreated PS}\ (g)}{\text{Glucan content/xylan and lignin contents of raw PS}\ (g)} \times 100\% \quad (6)$$

2.5. Characterization of PS

The crystallinity index (CrI) of cellulose within the pretreated PS was analyzed by X-ray diffractometer (XRD, Rigaku, Japan) subsequent to thorough grinding. The X-ray scans were conducted at a precise step size of 0.02°/s across a scattering angle range from 5° to 60°. Employing Eq. (7), the CrI was calculated according to the area of crystallization (I_{200}) and amorphous (I_{am}) region.

$$\text{CrI}\ (\%) = \frac{I_{200} - I_{am}}{I_{200}} \times 100\% \quad (7)$$

The accessibility of PS was assessed through a dyeing experiment with *Direct Red 28* dye. Specifically, 1 % PS was mixed with varying concentrations of dye solutions (0–4.0 g/L) in flasks and incubated in a constant temperature shaker at 60 °C and 150 rpm for 24 h. Following incubation, the suspension was centrifuged at 3500 *g* for 5 min, and the absorbance of the residual dye in the supernatant was measured at 498 nm using a UV spectrophotometer. The accessibility of PS was quantified by determining its maximum dye adsorption capacity, which was calculated by fitting the experimental data to the Langmuir isotherm model.

The surface area of lignin in PS was measured by an *Azure B* dye adsorption experiment. Various concentrations of dye were prepared in a sodium phosphate buffer solution with pH 7.0, after which 0.2 % (*w/v*) substrate suspension was introduced. The flasks were incubated at 25 °C and 150 rpm for 24 h. Upon completion, the mixture was centrifuged at 5000*g* for 5 min, and the absorbance of the resulting supernatant was measured at 647 nm using a UV spectrophotometer. The surface area of lignin in PS was subsequently calculated based on the Langmuir isotherm equation.

2.6. Isolation and characterization of lignin

The isolation of milled wood lignin (Björkman lignin) from raw and DES-pretreated PS was conducted for Nuclear Magnetic Resonance (NMR) analysis. Initially, 10 g of samples were subjected to ball milling for 48 h at 360 rpm using a planetary ball mill (JX-4G, Jingxin Industrial, China). The resulting powders were subsequently dispersed in a 96 % aqueous dioxane solution and stirred at 150 rpm for 24 h with this process repeated three times. The suspension was then centrifuged at 5000 *g* for 5 min to isolate the lignin extract. The isolated extract was rotary evaporated to yield a coarse lignin powder. Lignin purification was conducted sequentially using a 10 wt% dichloroethane/ethanol mixture (2:1, v/v), followed by ether treatment and air drying. The milled wood lignin in PS derived from CCL, CBL, DBL, and CAL pretreated PS were designated as L_{CCL}, L_{CBL}, L_{DBL}, and L_{CAL}, respectively. Finally, the lignin was characterized using a 600 MHz NMR spectrometer (Avance, Bruker, USA). For the 2D-HSQC analysis, 0.15 g lignin was dissolved in 0.5 mL DMSO-d_6. For the ^{31}P NMR analysis, 40 mg lignin was dissolved in a mixture (0.5 mL) of pyridine and $CDCl_3$ (1.6:1, v/v). Subsequently, *N*-hydroxy-5-norbornene-2,3-dicarboximide (internal standard) and TMDP (phosphating reagent) was sequentially added, and

the test was completed within 30 min. The resulting spectra was analyzed using Bruker software (version 4.1.4).

2.7. Quantum theory analysis of DES and DES:VG

The molecular structures of CTAB, CTAC, DTAB, ChCl and LA were optimized by Gaussian09 at M062X/6–311 + g(d,p) level [17]. The lignin model compound utilized in this study was veratrylglycerol-β-guaiacyl ether (VG), which contained *β*-O-4 ether linkages, aromatic hydroxyl groups, and two aliphatic hydroxyl groups. The cluster structures of various DES and their clusters formed with VG were constructed using MOPAC and Gaussian09 programs. For each cluster, 20 random cluster configurations were generated. These configurations were initially optimized semi-empirically at the PM6-DH$^+$ level, followed by further refinement using Gaussian09 at the B3LYP/6-31G* level to obtain detailed molecular electrostatic potential information. The electron density (ρBCP) and the Laplace of the electron density (∇^2ρBCP) were calculated for each DES and their clusters formed with VG. Ultimately, the lowest-energy configuration was selected from the 20 random configurations, and subsequent analyses were performed using the Multiwfn program, including molecular surface electrostatic potential calculations, atoms-in-molecules (AIM) analysis, and Independent Gradient Model with Hydrogen (IGMH) analysis.

3. Results and discussion

3.1. Property difference of DES

3.1.1. Kamlet-Taft solvatochromic parameters

To investigate the unique properties of DES, four distinct samples composed of ChCl:LA (CCL), CTAB:LA (CBL), DTAB:LA (DBL), and CTAC:LA (CAL) were prepared with a molar ratio of HBD to HBA set at 1:4. The multiparametric K-T parameters including the hydrogen bond acidity (*α*), hydrogen bond basicity (*β*), and polarizability (*π**), were utilized to quantitatively evaluate the polarity characteristics of these DES [30]. As presented in Table 1, alkylammonium-based DES have the lower *π** values compared to traditional CCL, specifically 0.83 for CBL, 0.59 for DBL, and 0.31 for CAL. Given that HBD was consistently comprised of an equal amount of LA, the predominant factors influencing the polarizability of these DES were attributed to the structural characteristics of HBA [31,32]. Specifically, the O in -OH group of the ChCl molecule possessed high electronegativity, whereas the C in -CH_3 group of the alkylammonium-based DES displayed relatively uniform electronegativity [33]. Additionally, due to the lower electronegativity of Br^- compared to Cl^-, the positive charge distribution in the CTAB molecule was more concentrated, resulting in higher polarity [34]. Notably, the DTAB with shorter alkyl chain had higher polarity than CTAB, likely due to differences in solubility and amphiphilicity [35]. Table 1 also presented the *β* parameter, which quantified the ability of DES to facilitate proton association through hydrogen bonding. The values were ranked as follows: CAL (1.08) > DBL (1.02) > CCL (0.74) > CBL (0.57). The capability of HBA to form hydrogen bonds was primarily characterized by its ability to donate electron density [36]. Specifically, the N in quaternary ammonium structures of both ChCl or alkylammonium halides could provide electron density, thereby facilitating hydrogen bond formation [37,38]. However, CTAB with longer alkyl chain allows C to contribute indistinctively to electron density [36]. Furthermore, the ammonium bromide exhibited greater flexibility compared to the ammonium chloride due to the smaller size of Br relative to Cl, leading to a stronger electron density for CTAC than CTAB [39]. Upon the formation of hydrogen bonds between HBA and protons from HBD, the electron density of DES became moderated, correlating closely with the *β* value of DES [40]. The parameter *α* was frequently employed to quantify the acidity of DES or its proton-donating capability [19]. As shown in Table 1, the *α* of CCL, CBL, DBL, and CAL were 1.38, 1.27, 1.32, and 1.17, respectively. Even with the identical HBD, DES displayed varying proton-supplying capacities, which were also closely related to the characteristics of HBA. Studies have shown that ChCl exhibited greater instability under acidic conditions compared to CTAB, negatively impacting its proton-providing capability associated with HBD [40]. Conversely, CTAB, with longer alkyl chain and acid instability, demonstrated adverse proton acceptance performance [41]. Moreover, CTAB was less likely to provide protons compared to CTAC due to the higher stability of Br^- relative to Cl^- [35]. In summary, the K-T parameter characteristics of DES showed a profound and intricate relationship with the structural attributes of HBA, consequently influencing the dissolution behavior of non-polar polymers. Among the four DES, the CBL with CTAB as HBA possessed the lowest *β* and highest *α* value, despite its *π** being marginally lower than that of CCL.

Table 1
The Kamlet-Taft parameter values for the DES with various HBA.

DES	DES abbreviation	Molar ratio	π*	α	β
ChCl:LA	CCL	1:4	0.89	1.38	0.74
CTAB:LA	CBL	1:4	0.83	1.27	0.57
DTAB:LA	DBL	1:4	0.59	1.32	1.02
CTAC:LA	CAL	1:4	0.31	1.17	1.08

3.1.2. Changes in structural features

The FTIR spectra of DES labeled with key functional groups were presented in Fig. 1a. The characteristic peak at 3445 cm^{-1} was attributed to the stretching vibration of -OH in CCL or -NH_3^+ in CBL, DEL, or CAL [42]. The broad peaks and significant vibration amplitude observed for CBL suggested the formation of hydrogen bonds between N and H within the CBL structure. The peaks at 2924 cm^{-1} and 2852 cm^{-1} corresponded to the characteristic stretching vibration of -CH_3 and -CH_2 in alkylammonium-based DES [21]. The peak at 1742 cm^{-1} was assigned to the C═O stretching vibration in LA, with its reduced intensity indicating weakened hydrogen bonding interactions due to HBD [5]. The C—N stretching vibrations were observed at 1637 cm^{-1} and 1127 cm^{-1}. The prominent peak areas of CBL and CAL may be attributed to the presence of internal covalent bonding within alkylammonium structure [43]. The peaks at 1473 cm^{-1} and 1403 cm^{-1} were attributed to in-plane and out-of-plane scissoring vibrations of the -CH_3 or -CH_2 in HBA, respectively. Notably, the out-of-plane scissoring peak intensity for -CH_3 and -CH_2 in CBL was significantly higher, which may be due to increased internal forces leading to enhanced vibrational amplitude [25]. The peaks at 1232 cm^{-1} and 949 cm^{-1} correspond to the stretching vibrations of C—O. The prominence of the C—O stretching vibration in DBL suggested that the O in this DES actively participated in hydrogen bond formation [44]. Based on the FTIR spectra analysis, it was reasonable to deduce that the N and O in the four DES likely serve as pivotal contributors to the establishment of enhanced hydrogen bonding networks. The ^{1}H NMR spectrum was used to further validate and refine the structure information of various DES. As illustrated in Fig. 1b, the peaks observed at 3.4 ppm was likely attributable to the proton signal arising from the internal hydrogen bond within the DES. The peaks around 4.8 ppm corresponded to the proton signals of the methine group in LA [15]. The peaks at 3.1 ppm, 2.5 ppm, 1.3 ppm, and 0.9 ppm were attributed to the methylene group bonded to the halogen element, the methyl group attached to the amino group, and the methylene group on the alkyl chain, respectively [32]. The peak at 4.1 ppm was attributed to the characteristic signal of hydrogen in the -COOH [45]. The appearance of proton peak associated with hydrogen bond indicated that alkylammonium halides were a viable strategy for preparing DES. Notably, the hydrogen bond peaks observed for DBL and CCL were more pronounced, potentially indicating stronger hydrogen bond interactions. Furthermore, TGA was employed to assess the thermostability of DES. As shown in Fig. 1c and d, CBL demonstrated superior thermal stability, with a mass loss of less than 10 % when heated to 186 °C. In contrast, the temperatures at which a 10 % mass loss occurred for CCL, DBL, and CAL

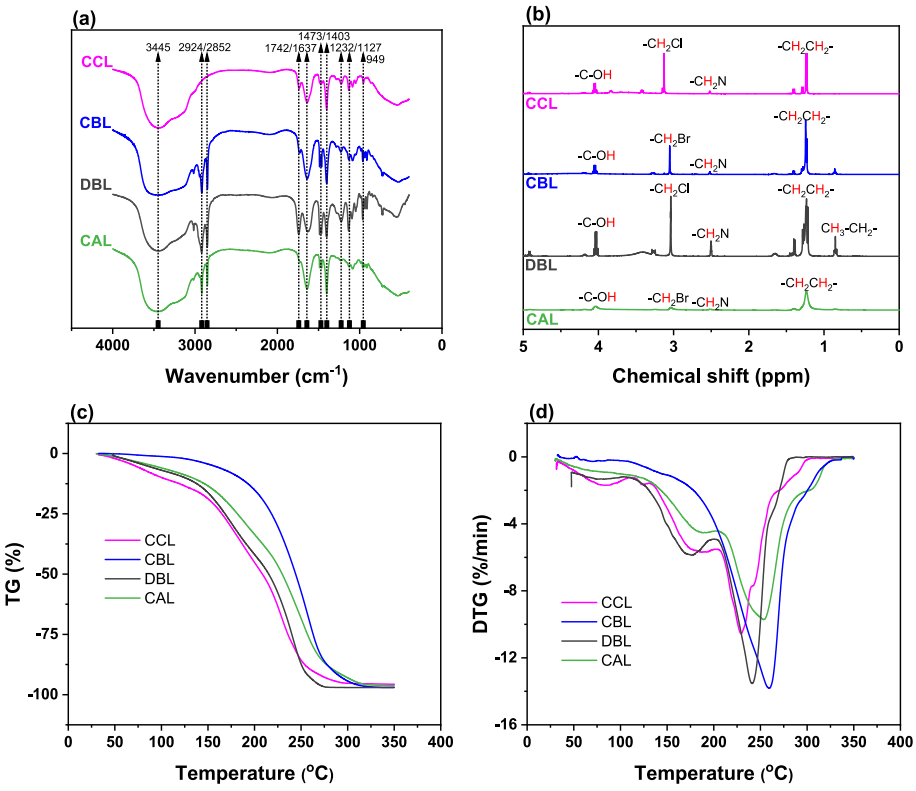


Fig. 1. The FTIR spectra (a), ^{1}H NMR spectra (b), TG spectra (c and d) of various DES.

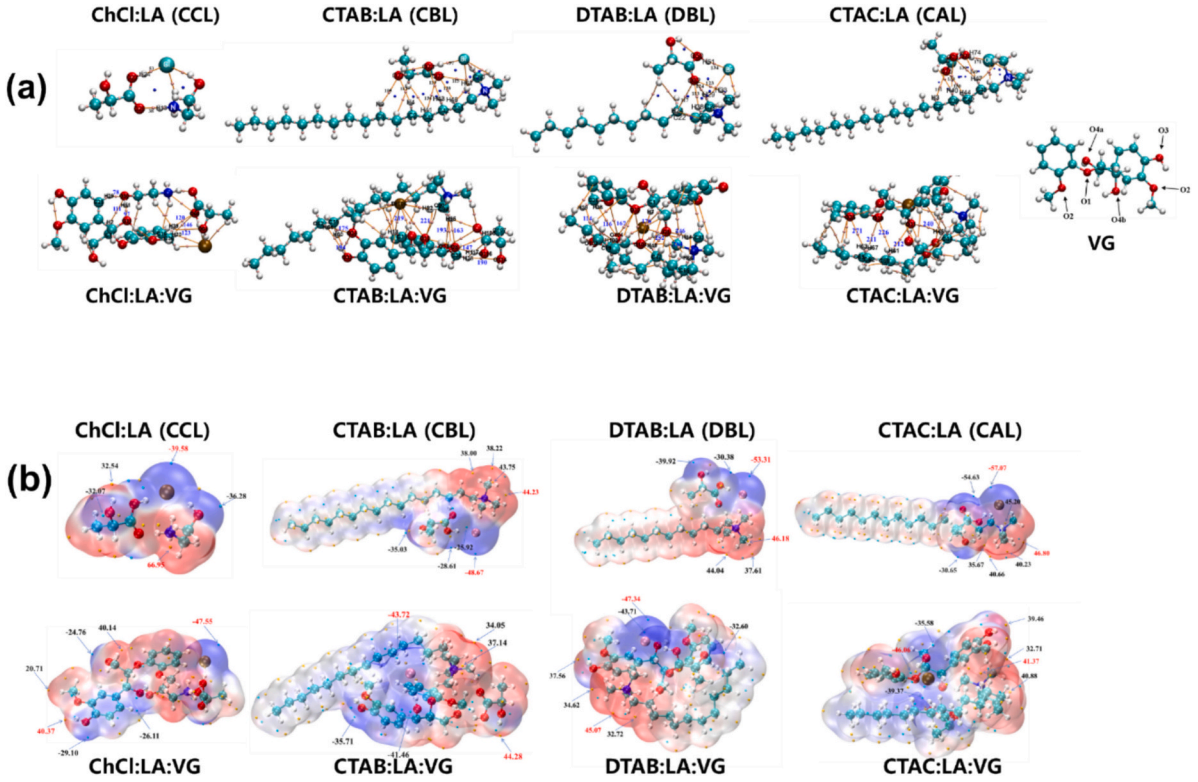


Fig. 2. AIM molecular graphs (a) and electrostatic potential coloring van der Waals surface diagram (b) of various DES and DES:VG.

were 113 °C, 135 °C, and 144 °C, respectively. The melting points of the four DES followed the order: CBL > CAL > DBL > CCL. Upon increasing the temperature to 267 °C, 298 °C, 307 °C, and 312 °C, the complete weight loss of the DES was observed in the sequence of DBL, CCL, CBL, and CAL, respectively. This observation aligned with previous reports, which highlighted that extended alkyl chains significantly augmented molecular interactions while impeding the free movement between molecules, thereby enhancing thermal stability [46,47]. Consequently, the DES prepared through the utilization of CTAB and CTAC demonstrated markedly superior thermal stability.

3.1.3. Quantum chemical calculation

Atoms in Molecules (AIM) theory was employed to investigate the intermolecular interactions within the DES [17]. Fig. 2a illustrated the bond critical points (BCP) associated with hydrogen bonds in the four DES, while Table S1 summarized the corresponding electron density (ρBCP) and Laplace function of the electron density (∇^2ρBCP). According to AIM theory, the standard range of ρBCP in hydrogen bonds was 0.002 to 0.035 a.u., and the ∇^2ρBCP values generally fell between 0.024 and 0.139 a.u. [15,44]. The ρBCP and ∇^2ρBCP values between HBA and HBD for all four DES were within these specified ranges, confirming that their predominant interaction mode was hydrogen bonding. Subsequently, the hydrogen bond energy (HBE) was calculated and categorized into four classes: weak (> − 2.5 kcal/mol), weak-medium (−2.5 to −14.0 kcal/mol), medium (−15.0 to −11.0 kcal/mol), and strong (<−15.0 kcal/mol) [44]. According to Table S1, the HBE between the halogen ions of CTAB, CTAC and DTAB with LA were − 7.656, −8.681 and − 9.848 kcal/mol, respectively. In contrast, the HBE between Cl in ChCl and LA was measured at −8.829 kcal/mol. These findings suggested that alkylammonium halide-based DES exhibited strong internal hydrogen bonding, with the most pronounced effect observed in short-chain brominated DTAB as HBA. Fig. 2b presented a van der Waals surface diagram illustrating the electrostatic potential coloring for various DES. Notably, there were negative electrostatic potentials near the Cl^- or Br^- of the HBA and the hydroxyl O of LA, while substantial positive potentials were observed near H and methyl groups. This confirmed that the pronounced negative potential surrounding functional group indicated their suitability as HBA, whereas the strong positive potential suggested their appropriateness as HBD [48]. The formation of hydrogen bonds between the HBD and HBA led to charge delocalization, thereby facilitating the formation of DES [49]. The electrostatic potential values of CCL, CBL, DBL, and CAL ranged from −39.58 to 66.95 kJ/mol, −48.67 to 44.23 kJ/mol, −53.31 to 46.18 kJ/mol, and − 57.07 to 46.80 kJ/mol, respectively. It was evident that the negative potential in CAL was more pronounced compared to the other DES, indicating a stronger interaction force between CTAC and LA within CAL. Additionally, the positive potential of CCL was more pronounced compared to the other three DES, likely due to multiple hydrogen bonds formed between the alkyl chains of the long-chain HBA and LA, as well as complementarity interactions between positive and negative regions [39]. These factors significantly diminished the energy and reduced the positive potential surrounding the valence group [47]. Fig. 3a illustrated the quantitative distribution of electrostatic potential for various DES, which aligned with the van der Waals surface diagrams. It was observed that DBL and CAL exhibited extensive charged regions, followed by CBL and CCL. This suggested that DTAB and CTAC have stronger interactions with DES formed by LA. Fig. 4a-4d presented the IGMH plots for various DES, serving as a computational method to visualize weak interactions. Within the isosurface formed between HBA and HBD, it was evident that the primary weak interactions occurred between the -OH in HBD and the Cl^- or Br^- of HBA. In the IGMH scatter plot of CCL, the density of blue scatter points was notably higher than that of green scatter points, indicating that hydrogen bonding was the predominant interaction between ChCl and LA. In the IGMH scatter plots of the other DES, the density in the green area was significantly greater, suggesting a strong van der Waals forces in CBL, DBL, and CAL. Furthermore, the blue region in DBL showed a higher density compared to that in CAL, with CBL exhibiting the lowest density. This observation also indicated that the DES prepared from the short-chain and brominated DTAB as well as CCL, demonstrated stronger hydrogen bonding, whereas the hydrogen bonding between the long-chain CTAB and LA was considerably weaker.

3.2. Principal chemical compositions of DES-pretreated PS

Table 2 depicted the glucan, xylan and lignin contents in DES-pretreated PS, as well as the separation degree (SD) of lignin (SD_L), xylan (SD_X), and both combined (SD_{LX}). Fig. 5 displayed the calculated removal of lignin and xylan, along with their direct correlation to SD. As shown in Table 2, the noncellulosic components of PS were significantly reduced following pretreatment with various DES. Notably, CBL-pretreated PS possessed the highest SD_L (1.4) and SD_{LX} (2.3), indicating its superior lignin depolymerization capability compared to commonly used CCL. This observation was corroborated by Fig. 5a, which showed that lignin removal for CBL-pretreated PS was the highest at 54.5 %, while CCL-pretreated PS demonstrated the highest xylan removal at 87.7 %. However, the DBL or CAL-pretreated PS achieved lignin removal of only 37.3 %–42.3 % and xylan removal of 69.3 %–71.7 %, respectively. Hence, DES prepared with long alkyl-chain and brominated CTAB as HBA possessed the best pretreatment performance.

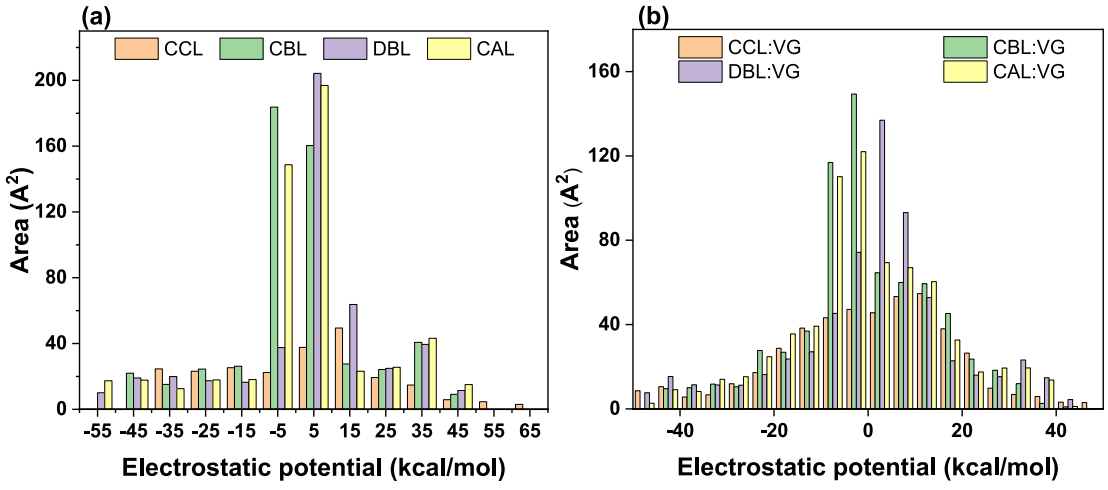


Fig. 3. The electrostatic potential quantitative distributions of the structurally optimized DES (a) and DES:VG (b).

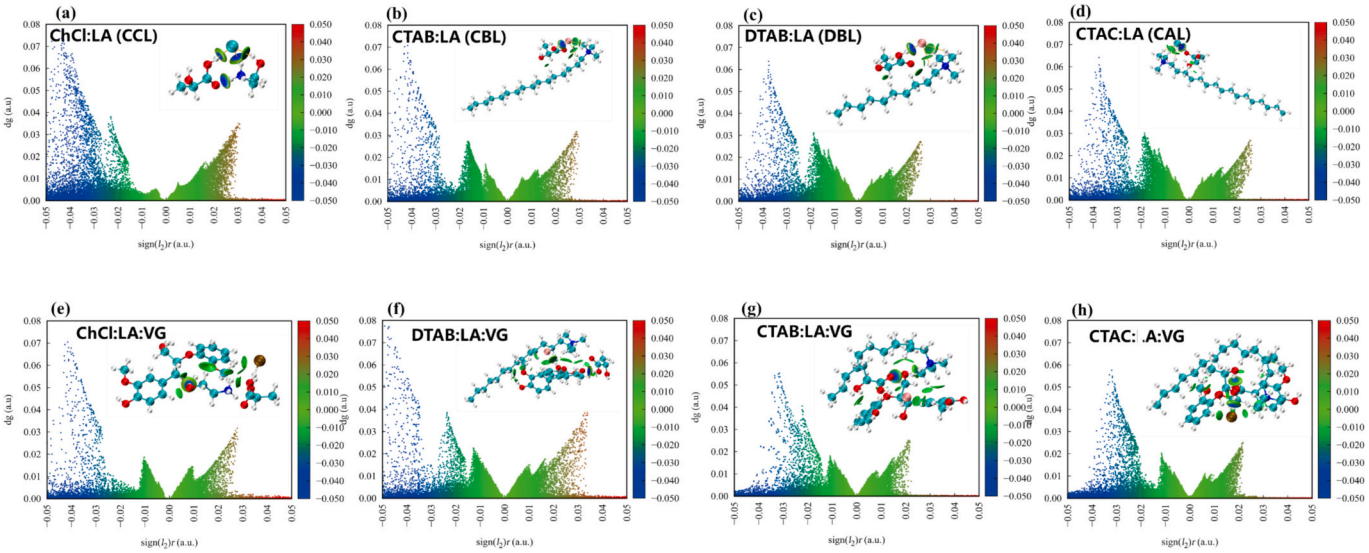


Fig. 4. IGMH plot and scatter plot of various DES (a-d) and DES:VG (e-h), blue represents hydrogen bonding force, green represents van der Waals force, and red represents intermolecular repulsive force. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
The chemical composition contents and separation degree of pretreated PS with various DES.

DES	DES abbreviation	Molar ratio	Temperature (°C)	Glucan (%)	Xylan (%)	Lignin (%)	SD_{LX}^{a}	SD_L	SD_X
–		–	–	40.9 ± 0.8	19.4 ± 0.2	31.6 ± 0.3	–	–	–
ChCl:LA	CCL	1:4	100	60.2 ± 0.2	4.4 ± 0.3	29.5 ± 1.2	2.2	1.1	5.5
CTAB:LA	CBL	1:4	100	59.0 ± 1.3	6.4 ± 0.2	25.2 ± 0.2	2.3	1.4	3.5
DTAB:LA	DBL	1:4	100	53.4 ± 1.5	8.7 ± 0.7	28.9 ± 0.3	1.8	1.1	2.5
CTAC:LA	CAL	1:4	100	51.4 ± 2.0	9.2 ± 0.6	30.5 ± 0.5	1.6	1.0	2.4
CTAB:LA		1:2	100	54.7 ± 1.7	10.2 ± 0.2	25.1 ± 0.3	1.9	1.4	2.1
		1:4	100	59.1 ± 1.5	6.7 ± 0.1	24.2 ± 0.8	2.4	1.4	3.3
		1:6	100	62.1 ± 1.6	4.8 ± 0.5	21.3 ± 1.0	3.0	1.6	4.7
		1:8	100	63.3 ± 2.1	2.8 ± 0.1	23.0 ± 0.8	3.1	1.5	8.2
		1:6	80	58.8 ± 1.0	9.7 ± 0.3	24.9 ± 0.3	2.1	1.5	2.3
			100	63.7 ± 0.5	5.7 ± 0.6	23.7 ± 1.0	2.7	1.5	4.1
			120	66.9 ± 0.9	3.5 ± 0.3	19.0 ± 0.6	3.7	1.9	6.7
			140	68.4 ± 2.2	2.3 ± 0.4	20.0 ± 0.7	3.8	1.8	10.2

a. SD_L, SD_X, and SD_{LX}, refer to the separation degree of lignin, xylan, and their combined from DES pretreated PS.

CBL characterized by robust van der Waals interactions and diminished hydrogen binding energy could significantly enhance the wettability and permeation of DES into hydrophobic lignin domains through their intrinsic surface-active properties, thereby promoting the efficient dissolution and extraction of lignin [26]. Furthermore, despite CTAB's relatively low negative potential, it effectively intensified local polarization via electron delocalization, giving rise to a potent electrostatic driving force that actively facilitated proton engagement in the depolymerization of lignin [28]. Subsequently, the parameters of the CBL pretreatment were meticulously optimized. Table 2 revealed that the lignin and xylan compositions of PS were further reduced as the molar mass of LA in CBL increased, leading to a gradual increase in SD_{LX} from 1.9 to 3.1. When the CBL pretreatment was conducted at a molar ratio of 1:6, the lignin removal of PS reached its peak at 61.7 %, while the highest xylan removal of 91.8 % was observed at a 1:8 M ratio of CBL, as shown in Fig. 5a. The increase in HBD content may enhance the mobility of DES and provide more protons for depolymerization [16]. In the temperature optimization, the lignin and xylan fractions in PS were significantly decreased as the temperature increased from 80 °C to 140 °C, with the SD_{LX} increasing from 2.1 to 3.8. Specifically, Fig. 5a displayed the removal of lignin and xylan rose from 56.3 % to 68.4 % and from 71.7 % to 94.0 %, respectively. The increased temperature likely facilitated the collapse of lignocellulosic structure, thereby enhancing the dissolution of DES [50]. Moreover, Fig. 5b also demonstrated that SD_X, SD_L, or SD_{LX} all showed a superior positive linear relationship ($y = ax^2 + bx + c$, $R^2 > 0.76$, $p < 0.05$) with lignin and xylan removal. These findings suggested that the enhanced removal of lignin and xylan was attributed to the depolymerization effect of DES, freed from systematic errors. The long alkyl-chain and brominated CTAB as HBA exhibited higher delignification efficiency compared to DTAB and CTAC. This study elucidated the significant potential of alkylammonium-based DES in pretreatment processes.

3.3. Cellulose digestibility of DES-pretreated PS

The cellulose digestibility of DES-pretreated PS after 72 h of enzymatic hydrolysis was evaluated and shown in Fig. 6. The CCL-pretreated PS achieved a cellulose digestibility of 57.8 %, whereas the CBL-pretreated PS reached 61.3 %. Clearly, the DES utilizing CTAB as HBA demonstrated greater effectiveness compared to conventional ChCl. However, the cellulose digestibility of CAL or DBL-pretreated PS was relatively lower at 46.2 % and 52.5 %, respectively. As discussed, CBL exhibited low hydrogen bond energy and high thermal stability, which facilitated the extraction and dissolution of non-cellulosic components. This characteristic likely contributed significantly to the enhanced enzymatic hydrolysis efficiency of PS. In subsequent optimization, it was observed that the cellulose digestibility of PS was gradually increased from 54.0 % to 72.7 % with decreasing CTAB molar mass (1:2–1:6) in CBL. This trend may be attributed to the influence of the molar ratio on the fluid characteristics of DES, which in turn affected the performance

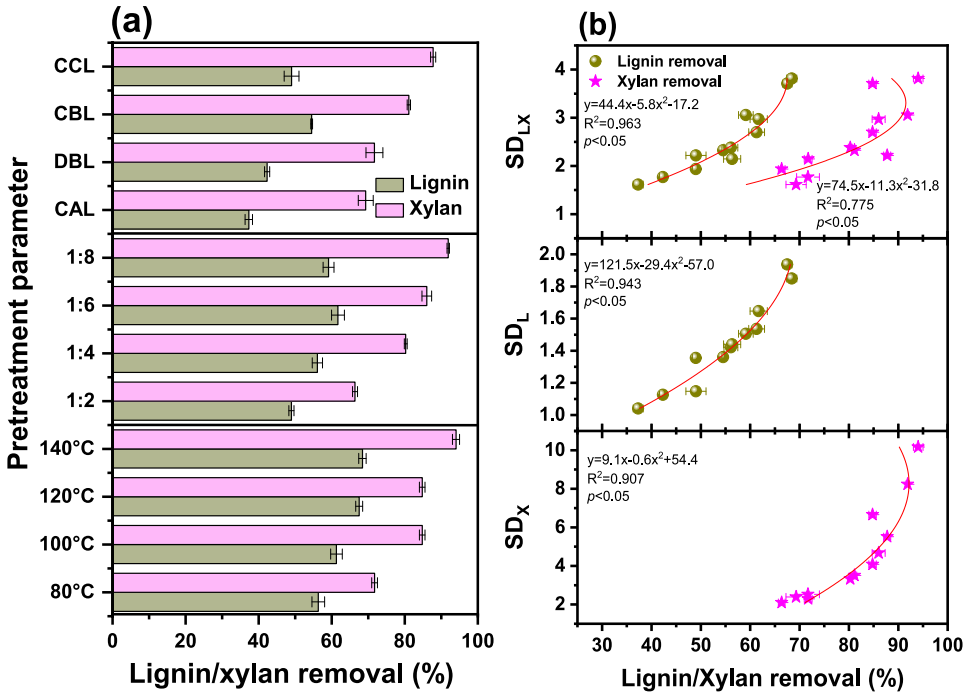


Fig. 5. The removal of lignin/xylan of the DES-pretreated PS (a), and its association with the separation degrees (b).

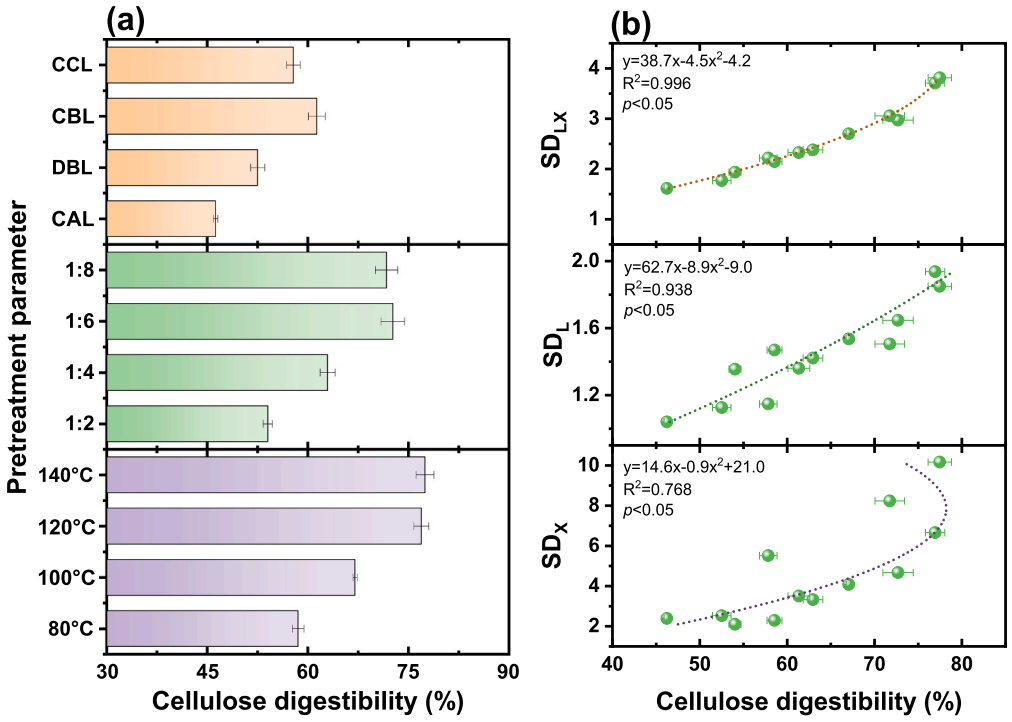


Fig. 6. The cellulose digestibility of the DES-pretreated PS in 72-h enzymatic hydrolysis (a), and its association with the separation degrees (b).

of CBL pretreatment. Fig. 6a also displayed that the cellulose digestibility of PS progressively increased from 58.5 % to 76.5 % as the pretreatment temperature increased from 80 °C to 140 °C. It was widely acknowledged that an appropriate temperature facilitated the provision of additional protons by DES, thereby enhancing the disruption of lignocellulose aggregation structures. Studies have reported that LA-based DES pretreatment with guanidine hydrochloride as the HBA increased delignification by 46.8 % and cellulose digestibility by 27.9 % compared to ChCl-based DES [51]. Liang observed that using tetrabutylammonium chloride as the HBA in DES achieved higher xylan and lignin removal, enhancing glucose yield by 26.8 % [19]. Here, it was determined that the HBA of DES were CTAB, ChCl, DTAB, and CTAC ranked according to the decreasing cellulose digestibility of pretreated PS. Fig. 6b showed positive linear relationships between the cellulose digestibility of PS and its SD_X, SD_L, or SD_{LX} values ($y = ax^2 + bx + c$, $R^2 > 0.77$, $p < 0.05$), confirming that the removal of lignin and xylan was the primary factor contributing to the enhanced cellulose digestibility. Compared with other DES, CBL showed superior delignification and

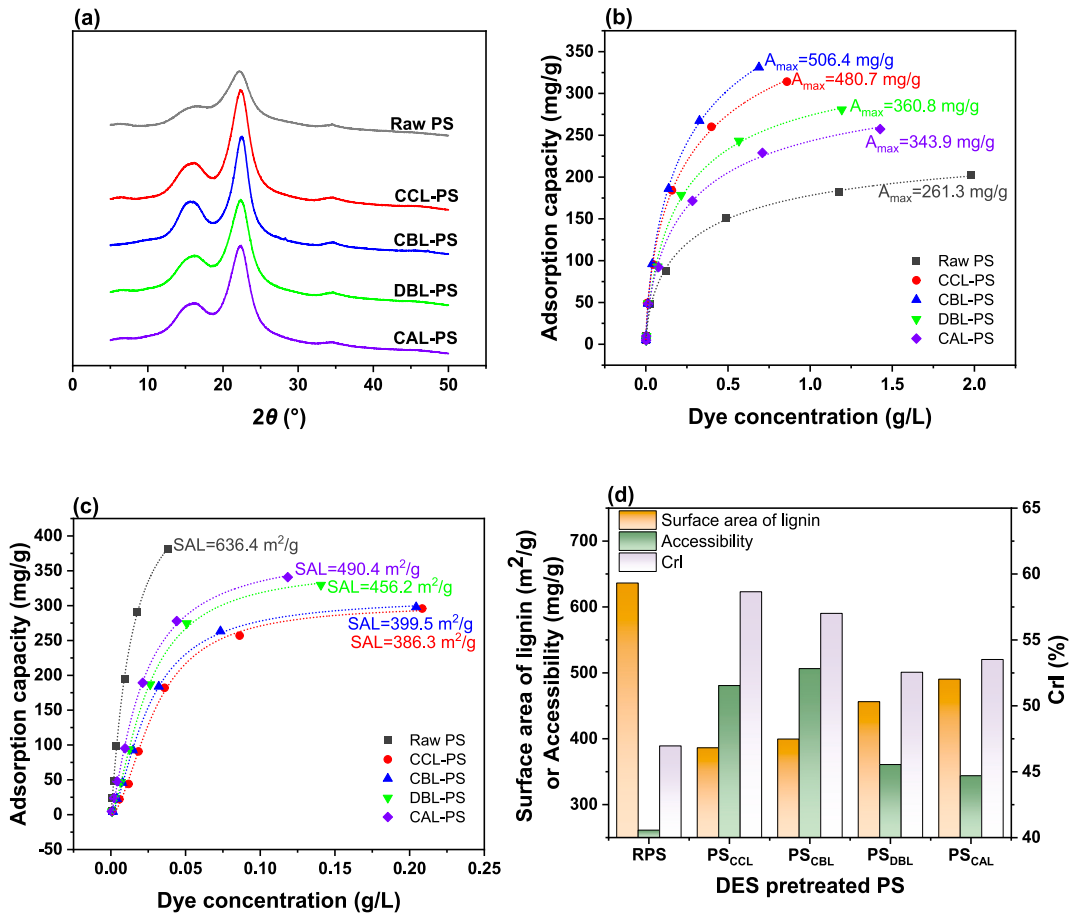


Fig. 7. The XRD pattern (a), Direct Red dye (b), Azure B dye adsorption pattern (c), and the value (d) of CrI, accessibility and lignin surface area of DES-pretreated PS. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
The crystallinity index (CrI), accessibility and lignin surface area of PS pretreated by CTAB:LA under different conditions.

DES	Molar ratio	Temperature (°C)	CrI (%)	Accessibility (mg/g)	Surface area of lignin (m^2/g)
CTAB: LA (CBL)	1:2	100	51.1	387.8	531.2
	1:6	100	57.0	506.4	399.5
	1:8	100	56.4	563.8	404.8
	1:6	80	54.7	465.2	486.8
	1:6	140	57.9	554.9	354.4

xylan removal capabilities, which memorably enhanced the bioconversion efficiency of PS.

3.4. Apparent characteristics of DES-pretreated PS

The CrI, accessibility, and surface area of lignin (SAL) of DES-pretreated PS were examined in this work. Fig. 7a showed the XRD patterns of raw and diverse DES-pretreated PS. The calculated CrI were presented in Fig. 7d, indicating that the CrI were increased from 46.9 % for raw PS to 58.7 %, 57.0 %, 52.5 %, and 53.1 % after CCL, CBL, DBL, or CAL pretreatment, respectively. This analysis revealed that both CCL and CBL pretreatments memorably increased the proportion of crystalline cellulose in PS. After various molar ratios and temperatures of CBL pretreatment, the CrI of PS were gradually increased from 51.1 % (1:2, 100 °C) to 57.9 % (1:6, 140 °C), as detailed in Table 3. Optimal conditions for CBL pretreatment could thus yield higher crystallinity in PS. Numerous studies have demonstrated that the increase in crystalline cellulose following lignocellulosic pretreatment was due to the reduction in amorphous components [10,52]. Consequently, the superior depolymerization capabilities of CBL pretreatment for lignin and xylan likely explained the observed elevation in CrI in PS. Furthermore, Fig. 7b illustrated the *Direct Red* dye adsorption capacity of the pretreated PS, where the adsorption maximum (A_{max}) was employed to quantify the affinity of PS for cellulase. As shown in Fig. 7d, the accessibility of raw PS was 261.3 mg/g, which was visibly increased to 480.7, 506.4, 360.8, and 343.9 mg/g after CCL, CBL, DBL, and CAL pretreatment, respectively. Table 3 displayed that the CBL pretreatment has further improved the accessibility of PS to 563.8 mg/g in the optimization processes. An increase in either the temperature or the molar ratio of LA in CBL positively influenced the accessibility of PS. Enhanced accessibility resulted from disrupting the lignocellulosic barrier by separating noncellulosic compositions, which improved enzyme adsorption on the substrate's cellulose [37]. Therefore, CBL pretreatment significantly removed lignin and xylan components, accelerating PS's accessibility and enhancing cellulose digestibility in this work. The SAL of PS was quantified using *Azure B* dye adsorption, as shown in Fig. 7c. The most substantial reduction in SAL was achieved through CCL pretreatment, which effectively reduced the SAL from 636.4 m^2/g for raw PS to 386.3 m^2/g (Fig. 7d). Comparable results were observed with CBL pretreatment, reducing the SAL to 399.5 m^2/g. After the molar ratio and temperature optimization of CBL pretreatment, the SAL of PS was further decreased from 531.2 to 354.4 m^2/g, respectively (Table 3). The pretreatment markedly reduced the residual lignin and caused the disintegration of the PS structure, thereby enhancing accessibility and crystallinity index (CrI). This may be the fundamental reason for the efficient hydrolysis observed in pretreated PS. Overall, the DES prepared

Table 4
The functional group contents, structural unit proportions and linkage contents of lignin in PS pretreated with various DES.

	L_R[a]	L_{CCL}	L_{CBL}	L_{DBL}	L_{CAL}
	Content (mmol/g)				
Aliphatic OH	7.46	3.43	4.11	5.08	5.33
Condensed phenolic OH	0.11	0.57	0.58	0.31	0.25
Non-condensed phenolic OH	0.25	0.49	0.51	0.29	0.27
Total phenolic OH	0.36	1.06	1.09	0.60	0.52
Carboxyl	0.08	0.48	0.52	0.29	0.25
	Content (1/100 Ar[b])				
β-O-4 (A)	58.4	41.3	39.8	50.9	53.6
β-β (B)	16.4	9.3	7.0	10.2	8.3
β-5 (C)	5.3	2.1	1.2	2.8	2.2
	Proportion (%)				
Syringyl (S)	50.1	55.6	57.7	53.3	51.8
Guaiacyl (G)	49.9	44.4	42.3	46.7	48.2
S/G	1.0	1.3	1.4	1.1	1.1

a. L_R, L_{CCL}, L_{CBL}, L_{DBL}, and L_{CAL} refer to the lignin in raw, ChCl:LA, CTAB:LA, DTAB:LA, and CTAC:LA pretreated PS.
b. Ar refers to the content of linkages per 100 aromatic units.

by CTAB demonstrated more pronounced and structurally meaningful effects on PS compared to those prepared using DTAB with short alkyl chains, CTAC containing alkylamine chloride, or ChCl lacking alkyl chains.

3.5. Mechanism of interaction between DES and lignin

3.5.1. Structural characteristics of lignin in DES-pretreated PS

The quantitative ^{31}P NMR technique was utilized to measure the contents of various hydroxyl and carboxyl functional groups in lignin from different DES-pretreated PS. As shown in Table 4, the initial aliphatic hydroxyl group content of L_R was measured at 7.46 mmol/g. Following pretreatment with CCL, CBL, DBL, and CAL, this value was decreased to 3.43, 4.11, 5.08, and 5.33 mmol/g, respectively. This indicated that the alkylamines, as HBA of DES, were consistent with CCL in their ability to promote the fragmentation and depolymerization of lignin during pretreatment, thereby facilitating the conversion of aliphatic hydroxyl groups to phenolic hydroxyl groups [53,54]. Notably, CBL showed a remarkable reduction than the DES prepared by chlorinated CTAC and short-chain DTAB, suggesting a stronger pretreatment intensity of CBL. Table 4 also displayed that the phenolic hydroxyl content of lignin in pretreated PS was advanced to diverse degrees. The condensed and non-condensed phenolic hydroxyl groups contents in L_{CCL}, L_{CBL}, L_{DBL}, and L_{CAL} were 0.57 and 0.49 mmol/g, 0.58 and 0.51 mmol/g, 0.31 and 0.29 mmol/g, 0.25 and 0.27 mmol/g, respectively. The elevated content of condensed hydroxyl groups in lignin feedbacked its fragmentation behavior in DES solvents, further elucidating the equivalence of pretreatment intensity between CBL and CCL [55]. The CBL pretreatment resulted in a higher number of hydroxyl groups in lignin than DBL and CAL, suggesting that the type of alkylamine influenced the effectiveness of DES pretreatment. The increased carboxyl content of lignin in pretreated PS further corroborated the superiority of CBL pretreatment in Table 4. Specifically, L_R contained a carboxyl group content of 0.08 mmol/g, while L_{CCL}, L_{CBL}, L_{DBL}, and L_{CAL} contained the carboxyl group contents of 0.48, 0.52, 0.29, and 0.25 mmol/g, respectively. The oxidative cracking of lignin typically resulted in a several-fold increase in carboxyl group content, thereby enhancing its solubility [5]. The CBL pretreatment notably elevated the carboxyl group content of lignin, signifying its superior pretreatment efficacy.

The relative abundances of the linkages in the aliphatic region (δ_C/δ_H

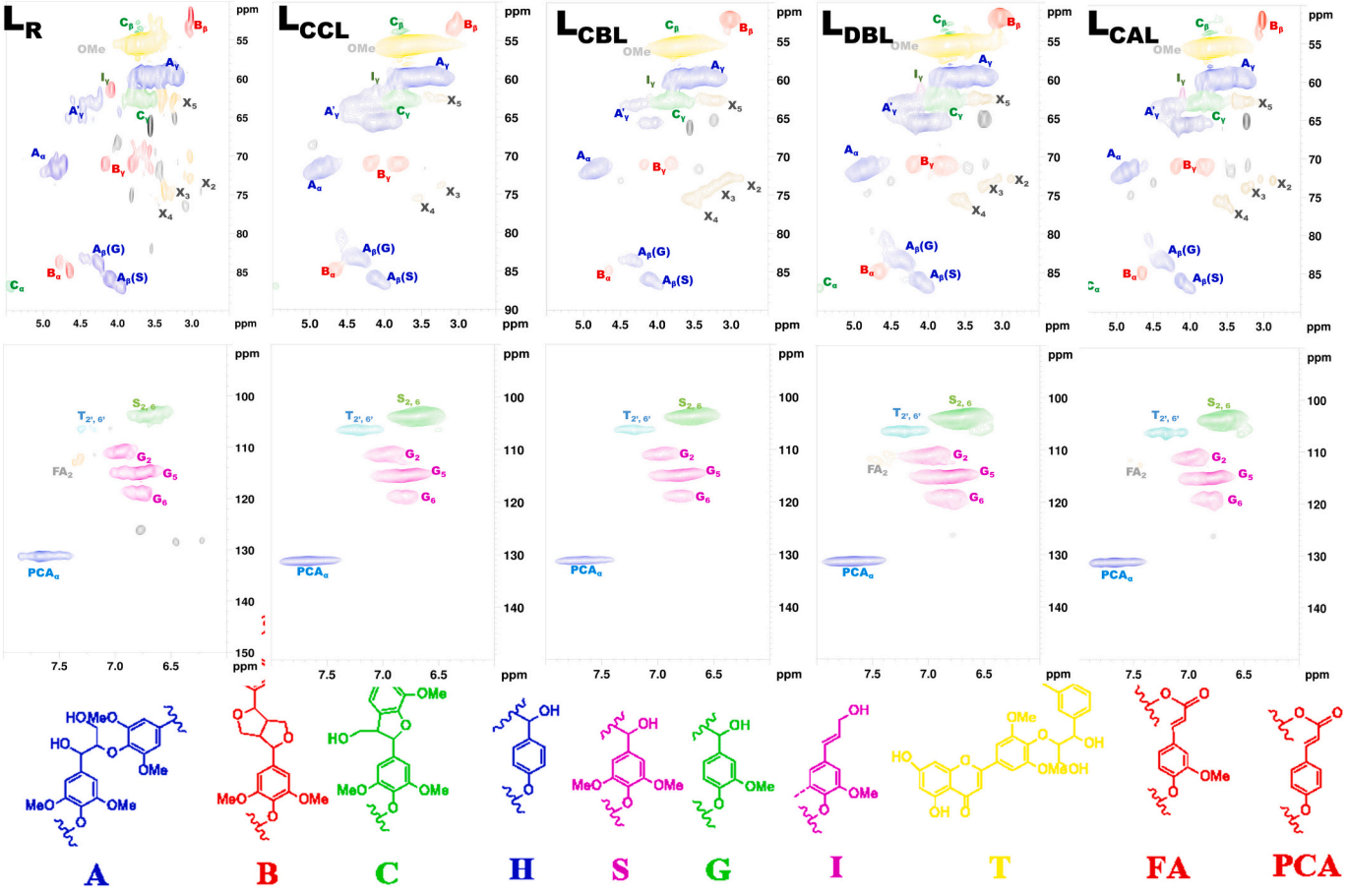


Fig. 8. The aliphatic regions and aromatic regions in the 2D-HSQC NMR spectra of MWL. The letter represents its structure code and shows in below of figure.

50.0–90.0/2.5–5.5) and the structural units in the aromatic region (δ_C/δ_H 90.0–150.0/6.0–8.0) were calculated based on 2D-HSQC NMR spectra (Fig. 8). It was observed that the predominant inter-unit linkages in L_R were *β*-O-4 (A), *β*-*β* (B) and *β*-5 (C), with syringyl (S) and guaiacyl (G) being the primary structural units. Following various DES pretreatment, the signal of linkages in PS's lignin were distinctly diminished, while the signal for structural units showed only minor changes. Table 4 depicted that the *β*-O-4, *β*-*β*, and *β*-5 content in L_R was 58.4/100 Ar, 16.4/100 Ar, and 5.3/100 Ar, respectively. After CBL pretreatment, the lignin linkages in PS exhibited the most obvious reduction, with values of 39.8/100 Ar, 7.0/100 Ar, and 1.2/100 Ar. The second most pronounced damage to lignin linkages was observed following CCL pretreatment, resulting in contents of 41.3/100 Ar for *β*-O-4, 9.3/100 Ar for *β*-*β*, and 2.1/100 Ar for *β*-5. These findings demonstrated that CTAB-based DES outperformed traditional CCL in terms of lignin dissolution, and a more pronounced depolymerization of the *β*-O-4 in lignin. The *β*-O-4, *β*-*β*, and *β*-5 content of L_{DBL} was 50.9/100 Ar, 10.2/100 Ar, and 2.8/100 Ar, while that of L_{CAL} was 53.6/100 Ar, 8.3/100 Ar, and 2.2/100 Ar. It was widely recognized that the acidic proton and internal hydrogen bonds of DES served as the primary forces responsible for disrupting lignin's ether and C—C bonds [18,50,56]. The fracture of the linkages during DES pretreatment constituted a prerequisite for lignin depolymerization and dissolution. In the aromatic region, the proportion of S units in lignin was notably elevated following DES pretreatment. As shown in Table 4, the proportion of S and G units in L_R was 50.1 % and 49.9 %, respectively. Following CCL, CBL, DBL and CAL pretreatments, the proportion of S units was increased to 55.6 %, 57.7 %, 53.3 % and 51.8 %, while the corresponding proportions of G units was reduced to 44.4 %, 42.3 %, 46.7 %, and 48.2 %. This transformation led to an increased S/G ratio of lignin in pretreated PS: 1.3 for L_{CCL}, 1.4 for L_{CBL}, 1.1 for L_{DBL}, and 1.1 for L_{CAL}. Previous studies have shown that the G unit of lignin with higher activity was more susceptible to oxidation than the S unit, thereby accelerating its degradation [21,57]. Meanwhile, the DES could supply protons to facilitate the demethoxylation of S units, converting them into G units [56]. These results collectively indicated that both CCL and CBL have profound effects on the lignin structure of PS, whereas DBL and CAL pretreatments exhibited weaker impacts. It has mentioned that the DES prepared with short-chain DTAB or chlorinated CTAC possessed higher hydrogen bond associations than CBL, but the depolymerization of lignin during pretreatment was inferior to that observed with CBL pretreatment. Hou et al. discovered that the hydrogen bond strength in DES was inversely proportional to their pretreatment efficiency, and the DES with higher hydrogen bond energy exhibited reduced lignin depolymerization [22]. Hence, as the hydrogen bond energy of DES decreased in the following order of CAL > DBL > CCL > CBL, the depolymerization and removal efficiency of lignin progressively increased in this work.

3.5.2. Quantum chemical calculation between DES and lignin model

The formation of a stable structure between DES and lignin is crucial for effective lignin dissolution during DES pretreatment. Therefore, the veratrylglycerol-β-guaiacyl ether (VG) was selected as a lignin model compound for quantum chemical calculations involving four DES systems [58]. Fig. 2a and b displayed the AIM molecular graphs and van der Waals surface maps with electrostatic potential coloring for four DES:VG clusters. In all clusters, the minimum electrostatic potential occurred near Cl^- or Br^-, with values of −47.55 kcal/mol (CCL-VG), −47.34 kcal/mol (DBL-VG), −46.06 kcal/mol (CAL-VG), and − 43.72 kcal/mol (CBL-VG). The lower absolute value of the electrostatic potential suggested weaker HBD-HBA interactions, which facilitated the formation of a hydrogen bond network between DES and VG. This stabilization enhanced the structural integrity of VG and improved lignin removal. Fig. 3b illustrated the quantitative distribution of electrostatic potential for four DES:VG clusters, aligning with the van der Waals surface diagram depicting electrostatic potential filling. It was evident that CBL:VG cluster exhibited a substantial area near an electrostatic potential of zero, followed by CAL:VG and DBL:VG, while CCL:VG displayed a notably smaller charged area. This suggested that CBL has wider interaction areas and stronger internal interactions with VG, consistent with its superior delignification performance. Subsequently, the intermolecular interactions were investigated using AIM theory. Fig. 3b showed the BCP representing the hydrogen bonds between the four DES:VG, while Table S1 summarized the ρBCP and ∇^2ρBCP. As shown in Fig. 3b and Table S1, the interaction sites in CCL:VG were relatively straightforward, primarily involving the alcohol hydroxyl group of VG. The ρBCP of the bond between the O of -OH and CCL was 0.0411, with a bond energy of −8.4564 kcal/mol, which fell within the covalent bond category. However, the molecular size of ChCl was considerably smaller than that of lignin, thereby restricting the effective interaction surface area and potential interactions between them. DBL established hydrogen bonds with both the ether bond and -OH of VG. Notably, the strongest interaction was observed between DTAB and the O of -OH in VG, characterized by a ρBCP value of 0.0222 and a bond energy of −4.2308 kcal/mol. In contrast, CAL formed relatively weak hydrogen bonds with both the ether bond and -OH of VG. Most prominently, CBL primarily interacted with the ether bond, alcohol hydroxyl group, and phenolic hydroxyl group of VG, resulting in the formation of multiple bond critical points. Notably, the strongest interaction was observed between LA and the O of the alcohol hydroxyl group of VG, characterized by a ρBCP value of 0.0510 and a bond energy of −10.6605 kcal/mol, which belonged to the covalent bond category [44]. This further suggested that CBL exhibited a more stable covalent bond structure with lignin compared to CCL, thereby accelerating the dissolution process of lignin. Fig. 4e-4h was the IGMH diagram, a computational method for visualizing weak interactions. The IGMH diagram corroborated the findings from the electrostatic potential surface diagram and AIM diagram. Specifically, the IGMH scatter plot of CBL:VG has a greater number and density of scatter points than the other three, with an additional peak of blue scatter points indicative of hydrogen bonding interactions. It could be inferred that the CBL:VG has stronger van der Waals force and hydrogen bonding interaction. The formation of stronger hydrogen bonds theoretically indicated that lignin rich in *β*-O-4 was more easily dissolved in CBL, and the quantitative results of lignin linkage bonds well confirmed this.

In conclusion, DES effectively broke the β-O-4 ether bonds and C—C bonds in lignin through the hydrogen bond system formed between alkylammonium and LA. Electrostatic potential mapping unveiled that the CBL and VG aggregates preferentially localize within regions of zero electric potential, suggesting an inverse relationship between hydrogen bond intensity and lignin's depolymerization efficacy. AIM further confirmed that strong covalent interactions formed between the -OH of CBL and VG, with a bond energy of −10.6605 kcal/mol, significantly higher than other types of DES, underscoring that the hydrogen bond network between CBL and lignin helped to effectively remove lignin. ^{31}P NMR analysis unveiled a marked decline in aliphatic OH content from 7.46 mmol/g to 4.11 mmol/g after CBL pretreatment, while the contents of phenolic OH and carboxyl groups exhibited a conspicuous rise. This intricate transformation paved the way for the establishment of new covalent linkages between DES and lignin, directly resulting in a reduction in the content of β-O-4 linkages from 58.4/100 Ar to 39.8/100 Ar. 2D-HSQC analysis further revealed that the G units were more vulnerable to oxidative degradation compared to S units in the L_R. The S/G ratio increased from 1.0 to 1.4 after CBL pretreatment, marking the most dramatic shift among all pretreatments. This increase was primarily driven by protons donated by DES, which actively catalyzed the cleavage of methoxy groups from lignin's aromatic framework. Notably, the negative correlation between hydrogen bond strength and the demethoxylation extent further validated the mechanism that weaker hydrogen bonds facilitated the formation of stronger covalent bonds between DES and lignin, thereby achieving the selective depolymerization of structural units. IGMH analysis unveiled a dense network of hydrogen bonds within the CBL system, complemented by an expansive

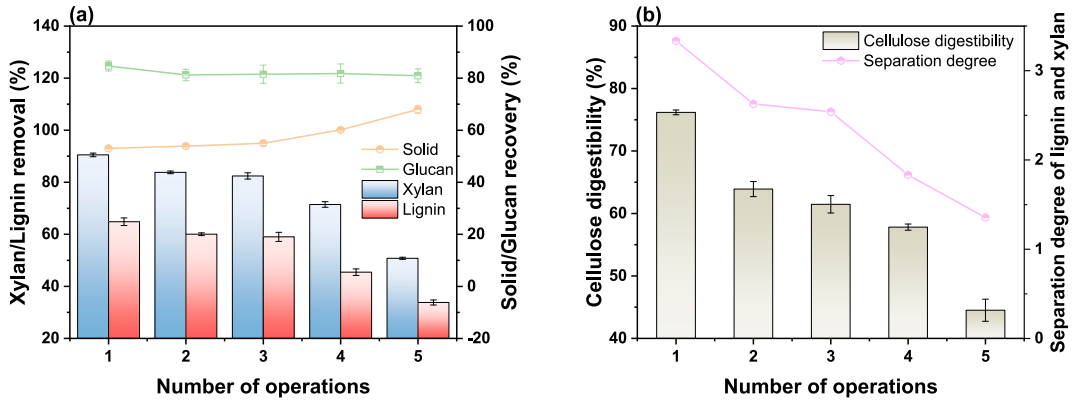


Fig. 9. The lignin/xylan removal and solid/glucan recovery (a), cellulose digestibility and separation degree (b) of the Pretreated PS with five cycles of CBL.

van der Waals interaction domain. This multi-acting site synergy enabled the simultaneous cleavage of ether linkages and -OH in VG, generating stable soluble precursors and ultimately driving efficient lignin deconstruction. Hence, although individual hydrogen bond energies in CBL were modest, it formed a more extensive and robust covalent interaction with lignin, which underpinned its superior performance of lignin removal and pretreatment efficiency.

3.6. Repeatability of DES in PS'S pretreatment

A key advantage of using DES for lignocellulosic fractionation is the recyclability of solvents, which determines its feasibility as a reaction solvent in biorefineries [59]. This work evaluated the recyclability of CBL (CTAB:LA) with the best pretreatment performance under optimal conditions (1:6, 120 °C). The results indicated that the changes in xylan and lignin content of PS progressively diminished as the number of CBL reuse cycles increased. As depicted in Fig. 9a, the recycled CBL exhibited consistently lower lignin and xylan removal efficiencies when applied to PS compared to fresh CBL pretreatment. After five cycles, the xylan removal decreased from 90.5 % to 50.8 %, while lignin removal dropped from 64.8 % to 33.8 %. Notably, the rate of decline was more pronounced in the last two cycles than in the first three. Fig. 9b showed that SD_{LX} also reflected this phenomenon, initially decreasing gradually from 3.3 to 2.5, and subsequently dropping sharply to 1.3. Concurrently, the varying removal efficiency of CBL pretreatment resulted in a progressive decline in the cellulose digestibility of PS from 76.2 % to 44.5 %. Lots of studies have demonstrated that the thermostability of recovered DES and its doped degradation products, which were derived from lignocellulose such as phenols, organic acids, or aldehydes, significantly affected their repeatability [16,51]. Xie et al. reported that the extraction efficiency of lignin from pine using benzyltrimethyl ammonium chloride:formic acid pretreatment decreased from 78 % to 67 % after five cycles [60]. It was also observed that the acidity of DES (ChCl:acetic acid) and thermostability of HBD were decreased by two cycles, resulting in a reduction of approximately 23 % in the enzymatic digestibility of sugarcane bagasse when it was reused for pretreatment [42]. Although the efficiency of CBL pretreatment declined by approximately 42 % following five consecutive cycles (the first two cycles only decreased by 19 %), it continued to demonstrate markedly superior performance and sustainability in comparison to conventional chemical pretreatment methods. Therefore, despite a moderate decrease in the reusability of DES, its considerable promise for large-scale industrial applications in biorefining remains firmly intact. In this work, the DES incorporating CTAB as HBA was selected to achieve effective fractionation and saccharification of PS, representing a notable innovation.

4. Conclusion

This work elucidated the advantages of diverse alkylammonium halide-based DES pretreatment in heightening cellulose digestibility of PS. The results demonstrated that the DES with CTAB as HBA possessed higher polarity, superior thermostability, and lower hydrogen bond energy compared to short-chain DTAB or chlorinated CTAC-based DES. The distinctive attributes of CBL ultimately resulted in the successful fractionation of 68.4 % lignin and 94.0 % xylan from PS, achieving a cellulose digestibility of 76.5 %. Characterization results indicated that the PS pretreated with CBL exhibited high accessibility and CrI, aligned with a reduced surface area of lignin. These findings underscored the effective disruption of the recalcitrant structure in PS by CBL. Quantum chemical calculations and NMR analysis revealed that the CBL disrupted the aromatic structure of lignin to a significantly greater extent compared to other DES, thereby demonstrating its exceptionally high efficacy. The proposal of innovative CBL broadened the category of DES and contributed to the sustainable and green advancement of biorefinery.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.seppur.2025.135641.

Data availability

No data was used for the research described in the article.

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